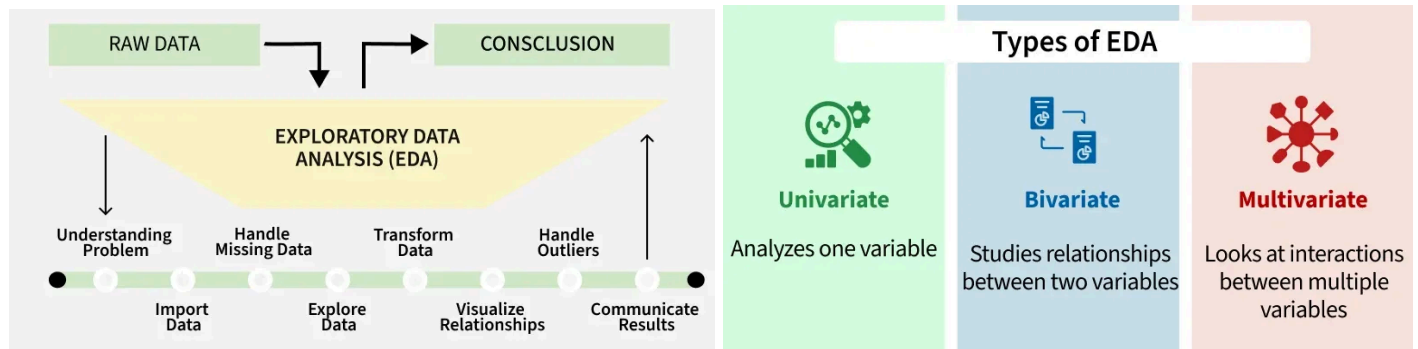


UNIT 1 INTRODUCTION TO EXPLORATORY DATA ANALYSIS

DEFINITION OF EDA

- Exploratory Data Analysis (EDA) is an initial, crucial step in data analysis to understand datasets by summarizing main characteristics, finding patterns, spotting anomalies, and checking assumptions, often using graphs and visualizations.
- Exploratory Data Analysis (EDA) is an approach to analyzing datasets to summarize their main characteristics, often using visual methods, before applying formal statistical modeling or hypothesis testing.
- EDA focuses on discovering patterns, detecting anomalies, testing assumptions, and understanding data structure.
- EDA was popularized by John W. Tukey, who emphasized using graphs and simple statistics to explore data.



COMPARISON OF EDA WITH CLASSICAL DATA SUMMARY MEASURES

Aspect	Exploratory Data Analysis (EDA)	Classical Data Summary Measures
Objective	To explore data, discover patterns, and gain insights	To summarize data using predefined statistics
Approach	Flexible, open-ended, and data-driven	Fixed, structured, and model-driven
Focus	Understanding data behavior and structure	Confirming assumptions and estimating parameters
Techniques Used	Graphical methods (histogram, boxplot, scatter plot, Q-Q plot)	Numerical measures (mean, median, variance, standard deviation)

Assumptions	Minimal or no assumptions about data distribution	Strong assumptions such as normality and independence
Nature of Analysis	Exploratory	Confirmatory
Handling Outliers	Identifies and highlights outliers	Often hides outliers within averages
Visualization	Heavy use of visualizations	Little or no visualization
Adaptability	Easily adapts to different types of data	Limited flexibility
Stage of Use	Used at the initial stage of data analysis	Used after data is assumed to fit a model
Result Interpretation	Intuitive and visual	Mathematical and statistical
Error Detection	Helps detect data errors and inconsistencies	Less effective in detecting data issues

GOALS OF EDA

- **Understand the Data Distribution**
- EDA helps in understanding how data values are distributed across a dataset. By using tools like histograms, density plots, and summary statistics, analysts can identify whether the data follows a normal, skewed, or uniform distribution. Understanding distribution is essential for selecting appropriate statistical methods and models.
- **Identify Outliers and Anomalies**
- One of the key goals of EDA is to detect unusual observations that deviate significantly from the rest of the data. These outliers may result from data entry errors, measurement issues, or genuine extreme events. Techniques such as box plots and scatter plots are commonly used to identify such anomalies.
- **Detect Patterns, Trends, and Relationships**
- EDA helps uncover hidden patterns and trends within the data. Scatter plots, correlation matrices, and line graphs are used to study relationships between variables. Identifying these relationships is important for understanding dependencies and for building predictive models.
- **Check Data Quality Issues**
- EDA assists in identifying data quality problems such as missing values, duplicate records, inconsistent formats, and incorrect entries. Early detection of these issues allows proper data cleaning and preprocessing, which improves the reliability of analysis results.
- **Validate Assumptions Before Applying Statistical Models**
- Many statistical and machine learning models rely on assumptions like normality, linearity, independence, and homoscedasticity. EDA helps test these assumptions using visual and numerical methods before model building, reducing the risk of incorrect conclusions.

- **Guide Feature Selection for Modeling**
- EDA helps determine which variables are relevant and influential for analysis or prediction. By analyzing correlations, variance, and importance of features, irrelevant or redundant variables can be removed. This leads to simpler, more accurate, and efficient models.

UNDERLYING ASSUMPTIONS IN EDA

- Although Exploratory Data Analysis is a flexible and open-ended approach, it is based on certain practical assumptions about real-world data. These assumptions guide how data should be explored and interpreted.
- EDA assumes that data should be observed, explored, and understood first, rather than being forced into predefined statistical models or assumptions. This makes EDA a data-driven and discovery-oriented approach.

1. Data May Contain Outliers or Noise

- EDA assumes that real-world data is rarely perfect and often includes outliers, extreme values, or random noise. Instead of immediately removing such values, EDA encourages identifying and studying them to understand whether they represent errors, rare events, or important information. Tools like box plots and scatter plots are commonly used for this purpose.

2. Data Distributions May Not Follow Normality

- Unlike classical statistical methods, EDA does not assume that data follows a normal (Gaussian) distribution. Data may be skewed, multi-modal, or irregular in shape. EDA techniques such as histograms, density plots, and Q-Q plots help visualize the true distribution without imposing strict assumptions.

3. Relationships Between Variables May Be Non-Linear

- EDA assumes that relationships among variables are not always linear. In many real-world datasets, variables may show complex or non-linear relationships. Scatter plots and pair plots are used in EDA to visually explore these relationships before selecting suitable analytical or modeling techniques.

4. Real-World Data Is Often Imperfect and Incomplete

- EDA recognizes that datasets frequently contain missing values, inconsistencies, duplicate records, or measurement errors. Instead of ignoring these issues, EDA focuses on detecting and understanding them early so that appropriate data cleaning and preprocessing steps can be applied.

IMPORTANCE OF EDA

- EDA helps to understand the dataset by showing how many features it has, what type of data each feature contains and how the data is distributed.
- It helps to identify hidden patterns and relationships between different data points which help us in model building.
- It allows us to identify errors or unusual data points (outliers) that could affect our results.
- The insights gained from EDA help us to identify the most important features for building models and guide us on how to prepare them for better performance.
- By understanding the data it helps us in choosing best modeling techniques and adjusting them for better results.

INTRODUCTION TO DIFFERENT TECHNIQUES TO TEST THE ASSUMPTIONS INVOLVED IN EDA

- In Exploratory Data Analysis, various graphical and statistical techniques are used to examine whether the data satisfies basic assumptions such as normality, linearity, presence of outliers, independence, and constant variance. These techniques help analysts understand data behavior before applying formal statistical or machine learning models.

1. Graphical Techniques

- **Histogram**
- Histograms display the frequency distribution of data values and help assess the shape of the data distribution. They are used to check whether the data is normally distributed, skewed, or multi-modal.
- Assumption Tested: Normality, Skewness
- **Box Plot**
- Box plots summarize data using quartiles and highlight outliers. They help evaluate data spread and symmetry.
- Assumption Tested: Presence of outliers, Data variability
- **Scatter Plot**
- Scatter plots show the relationship between two numerical variables and are useful for identifying linearity or non-linearity.
- Assumption Tested: Linearity, Correlation
- **Q-Q Plot (Quantile-Quantile Plot)**
- Q-Q plots compare the quantiles of the data with a theoretical distribution (usually normal). If points fall on a straight line, the data follows that distribution.
- Assumption Tested: Normality
- **Density Plot**

- Density plots provide a smooth representation of the data distribution, helping to observe distribution shape and peaks.
- Assumption Tested: Distribution pattern, Data smoothness

2. Statistical / Numerical Techniques

- **Measures of Central Tendency**
- Mean, median, and mode help understand where the data is centered. Differences between mean and median indicate skewness.
- Assumption Tested: Symmetry of distribution
- **Measures of Dispersion**
- Variance, standard deviation, and range indicate how spread out the data is.
- Assumption Tested: Data variability, Consistency
- **Skewness and Kurtosis**
- Skewness measures asymmetry, while kurtosis measures the peakedness of the distribution.
- Assumption Tested: Normality, Distribution shape
- **Correlation Analysis**
- Correlation coefficients measure the strength and direction of relationships between variables.
- Assumption Tested: Strength of association, Linearity
- **Missing Value and Data Quality Checks**
- Missing value counts
- Duplicate detection
- Inconsistent value checks
- Assumption Tested: Completeness and reliability of data

ROLE OF GRAPHICS IN DATA EXPLORATION

- Graphics play a central role in Exploratory Data Analysis (EDA) by enabling analysts to visually examine data and quickly understand its underlying structure. Visual representations make complex datasets easier to interpret and reveal patterns that are often hidden in numerical summaries.
- **Understanding Data Distribution**
- Graphs such as histograms and density plots help in visualizing how data values are distributed. They reveal the shape of the distribution (normal, skewed, or multimodal), central tendency, and spread, which are difficult to infer from numerical measures alone.
- **Identification of Outliers and Anomalies**
- Box plots and scatter plots make it easy to detect outliers and unusual observations. These graphical tools highlight extreme values that may represent data errors or important rare events, allowing analysts to investigate them further.
- **Detecting Patterns and Trends**

- Line charts and scatter plots help identify trends, clusters, and patterns over time or across variables. Graphics enable quick recognition of upward or downward trends and recurring patterns in data.
- **Studying Relationships Between Variables**
- Graphics such as scatter plots, pair plots, and heatmaps help analyze relationships and dependencies between variables. They reveal correlation, linearity, non-linearity, and interaction effects more intuitively than numerical correlation coefficients alone.
- **Validating Statistical Assumptions**
- Graphs such as Q-Q plots and residual plots help test assumptions like normality, linearity, and constant variance. Visual inspection allows early detection of assumption violations before applying statistical or machine learning models.
- **Improving Data Quality**
- Visualization helps identify missing values, inconsistencies, and data entry errors. Sudden gaps, unexpected spikes, or irregular patterns in graphs often indicate data quality problems.
- **Supporting Decision Making and Communication**
- Graphics simplify complex data findings and make them accessible to both technical and non-technical audiences. Clear visualizations support better interpretation, communication, and data-driven decision-making.

INTRODUCTION TO SINGLE, BI AND MULTIDIMENSIONAL GRAPHICAL REPRESENTATION

- Graphical representations are a fundamental part of Exploratory Data Analysis (EDA). Based on the number of variables involved, graphical methods are classified into single-dimensional (univariate), bi-dimensional (bivariate), and multidimensional (multivariate) representations. These visualizations help in understanding data behavior, relationships, and structure.

1. Single-Dimensional (Univariate) Graphical Representation

- Single-dimensional graphs represent only one variable at a time. They focus on understanding the distribution and characteristics of individual variables.
- Purpose:
 - Understand data distribution
 - Identify central tendency and spread
 - Detect outliers
- Common Graphs:
 - Histogram
 - Box Plot
 - Density Plot
 - Bar Chart (categorical data)

- Example:
- Distribution of student marks in an exam.
- Key Insight:
- Shows how values are spread within a single variable.

2. Bi-Dimensional (Bivariate) Graphical Representation

- Bi-dimensional graphs involve two variables and are used to study relationships and dependencies between them.
- Purpose:
- Identify correlation or association
- Check linear or non-linear relationships
- Compare two variables
- Common Graphs:
- Scatter Plot
- Line Graph
- Grouped Bar Chart
- Paired Box Plot
- Example:
- Study hours vs exam scores.
- Key Insight:
- Shows how one variable changes with another.

3. Multidimensional (Multivariate) Graphical Representation

- Multidimensional graphs represent three or more variables simultaneously. These are used for advanced data exploration to identify complex relationships and patterns.
- Purpose:
- Analyze interactions among multiple variables
- Detect clusters and patterns
- Reduce data complexity visually
- Common Graphs:
- 3D Scatter Plot
- Heatmap
- Pair Plot
- Bubble Chart
- Parallel Coordinates Plot
- Example:
- Customer data with age, income, and spending score.
- Key Insight:
- Reveals hidden structures and interactions in high-dimensional data.

DATA PREPARATION, CONSISTENCY AND POLLUTION OF DATA

- Before applying Exploratory Data Analysis or building models, data must be properly prepared. The quality of analysis largely depends on how clean, consistent, and reliable the data is.

1. Data Preparation

- Data Preparation is the process of converting raw data into a clean, structured, and usable format for analysis.
- Key Steps in Data Preparation:
 - Data collection from multiple sources
 - Handling missing values (removal or imputation)
 - Removing duplicate records
 - Correcting data entry errors
 - Data transformation (normalization, scaling, encoding)
 - Formatting and type conversion
- Importance:
 - Improves data quality
 - Reduces errors in analysis
 - Increases model accuracy
- Example:
 - Converting date formats from different sources into a single standard format.

2. Data Consistency

- Data Consistency refers to the uniformity and reliability of data across the dataset and across different sources.
- Causes of Inconsistency:
 - Different units of measurement (kg vs pounds)
 - Multiple formats for the same data (Male/M, Yes/1)
 - Conflicting values for the same record
 - Data collected from heterogeneous systems
- Ensuring Data Consistency:
 - Standardizing units and formats
 - Applying validation rules
 - Using reference dictionaries
 - Cross-checking data across sources
- Importance:
 - Ensures correctness and comparability

- Prevents misleading analysis
- Maintains data integrity

3. Data Pollution

- Data Pollution refers to the presence of inaccurate, irrelevant, or misleading data that contaminates the dataset and degrades analysis quality.
- Sources of Data Pollution:
 - Noise and random errors
 - Outliers and extreme values
 - Missing or incomplete data
 - Duplicate and irrelevant records
 - Incorrect data entry
- Effects of Data Pollution:
 - Biased or incorrect results
 - Reduced model performance
 - Poor decision-making
- Handling Data Pollution:
 - Outlier detection and treatment
 - Noise filtering
 - Removing irrelevant features
 - Data cleaning and validation

OUTLIERS AND LEVERAGE DATA

- In Exploratory Data Analysis (EDA), identifying unusual observations is crucial because they can significantly influence analysis results and model performance. Outliers and leverage points are two important concepts related to unusual data values.
- **Outliers**
- An outlier is a data point that lies significantly far away from the majority of observations in a dataset.
- **Causes of Outliers:**
 - Data entry or measurement errors
 - Sampling errors
 - Natural variability in data
 - Rare but valid events
- **Types of Outliers:**
 - Univariate outliers – Extreme values in a single variable
 - Multivariate outliers – Unusual combinations of values across multiple variables
- **Detection Techniques:**
 - Box plots

- Z-score method
- Interquartile Range (IQR) method
- Scatter plots
- **Impact of Outliers:**
- Distort mean and standard deviation
- Affect regression models
- Lead to misleading conclusions
- **Leverage Data (Leverage Points)**
- Leverage points are observations with extreme values of independent (input) variables. These points may or may not have unusual output values.
- **Key Characteristics:**
- Located far from the center of predictor variables
- Have the potential to strongly influence regression models
- Not all leverage points are outliers
- **Detection Techniques:**
- Scatter plots of independent variables
- Hat matrix values
- Cook's distance
- **Impact of Leverage Points:**
- Can disproportionately influence regression line
- May hide or exaggerate relationships
- Affect model stability

NOISY DATA

- Noisy data refers to data that contains random errors, irrelevant variations, or distortions that do not represent true underlying patterns. Noise obscures meaningful information and reduces the quality of data analysis.
- **Sources of Noisy Data:**
- Measurement or sensor errors
- Data entry mistakes
- Transmission errors
- Environmental interference
- Incomplete or imprecise data collection
- **Characteristics of Noisy Data:**
- Random and unpredictable variations
- Reduces signal clarity
- Makes patterns harder to detect
- Increases variance without adding information
- Impact of Noisy Data on Analysis
- Misleading trends and patterns

- Reduced accuracy of statistical models
- Poor machine learning performance
- Increased false conclusions
- **Detection of Noisy Data:**
- Scatter plots showing random dispersion
- Box plots with irregular spread
- Residual plots with no clear pattern
- High variance or instability in measurements
- **Handling and Reduction of Noisy Data**
- 1. Data Smoothing
- Moving averages
- Bin means or medians
- 2. Filtering Techniques
- Noise filters (e.g., low-pass filters)
- Signal processing methods
- 3. Outlier Detection and Treatment
- Removing or correcting extreme noise values
- 4. Data Aggregation
- Combining data to reduce random variation
- 5. Data Cleaning
- Correcting errors
- Removing irrelevant records

MISSING VALUES

- Missing values occur when no data value is stored for a variable in an observation. They appear as blanks, nulls, NaN, or special symbols and can significantly affect data analysis if not handled properly.

Causes of Missing Values

- Data entry errors
- Non-response in surveys
- Sensor or system failures
- Data corruption during transfer

- Incomplete data collection

Types of Missing Values

1. Missing Completely at Random (MCAR)

- The missingness is independent of both observed and unobserved data.
- Example: Random loss of survey forms.

2. Missing at Random (MAR)

- The missingness depends on observed data but not on the missing value itself.
- Example: Income missing mostly for younger respondents.

3. Missing Not at Random (MNAR)

- The missingness depends on the unobserved value itself.
- Example: People with very high income not reporting income.

Impact of Missing Values

- Biased analysis results
- Reduced statistical power
- Incorrect model training
- Misleading conclusions

Detection of Missing Values

- Counting null or NaN values
- Data summary statistics
- Visual inspection using heatmaps or tables

Handling Missing Values

1. Deletion Methods

- Listwise deletion: Remove rows with missing values
- Pairwise deletion: Use available data only

2. Imputation Methods

- Mean, median, or mode substitution
- Forward or backward fill
- Regression-based imputation
- K-nearest neighbors (KNN) imputation

3. Advanced Methods

- Multiple imputation
- Model-based techniques

IMPUTATION OF MISSING AND EMPTY PLACES WITH DIFFERENT TECHNIQUES

- Imputation is the process of replacing missing or empty data values with meaningful substitutes so that the dataset remains complete and suitable for analysis. Proper imputation improves data quality and prevents bias in statistical analysis and machine learning models.

1. Simple (Statistical) Imputation Techniques

1. Mean Imputation

- Missing values are replaced with the mean of the available values in that variable.

Used when:

- Data is numerical
- Distribution is approximately normal

Advantage: Simple and fast

Limitation: Reduces data variability

2. Median Imputation

- Missing values are replaced with the median of the variable.

Used when:

- Data is skewed
- Presence of outliers

Advantage: Robust to outliers

Limitation: Ignores relationships with other variables

3. Mode Imputation

- Missing values are replaced with the most frequent value.

Used when:

- Data is categorical

Advantage: Maintains category consistency

Limitation: Can introduce bias if overused

2. Deletion-Based Techniques (Alternative to Imputation)

1. Listwise Deletion

- Entire records containing missing values are removed.

Advantage: Simple

Limitation: Loss of data

2. Pairwise Deletion

- Uses all available data without deleting entire rows.

Advantage: Preserves more data

Limitation: Inconsistent sample size

3. Time-Series Imputation Techniques

1. Forward Fill (FFill)

- The missing value is replaced by the previous known value.

2. Backward Fill (BFill)

- The missing value is replaced by the next known value.

Used when:

- Time-ordered data

4. Predictive and Model-Based Imputation Techniques

1. Regression Imputation

- Predicts missing values using regression models based on other variables.

Advantage: Uses relationships between variables

Limitation: Can overfit

2. K-Nearest Neighbors (KNN) Imputation

- Uses values from similar data points to fill missing values.

Advantage: Preserves data patterns

Limitation: Computationally expensive

3. Multiple Imputation

- Generates multiple plausible values and combines results.

Advantage: Statistically robust

Limitation: Complex to implement

5. Advanced and Domain-Specific Techniques

- Expectation-Maximization (EM)
- Machine learning-based imputation
- Domain-rule imputation

MISSING PATTERN AND ITS IMPORTANCE

- A missing pattern refers to the structure, distribution, and behavior of missing values in a dataset. It describes where, how often, and in what way data values are missing across variables and observations.
- Understanding missing patterns helps determine whether missing data is random or systematic and guides the selection of appropriate imputation techniques.

Types of Missing Patterns

1. Missing Completely at Random (MCAR)

- Missing values occur randomly and are unrelated to any variable.
- Example: Random data loss due to system failure.
- Pattern: No specific structure.

2. Missing at Random (MAR)

- Missing values depend on observed variables but not on the missing values themselves.
- Example: Income missing more frequently for younger individuals.
- Pattern: Structured but explainable using other variables.

3. Missing Not at Random (MNAR)

- Missing values depend on the unobserved value itself.
- Example: High-income individuals not disclosing income.
- Pattern: Systematic and biased.

Methods to Identify Missing Patterns

- Missing value summary tables
- Heatmaps or matrix plots
- Percentage of missing values per variable
- Pairwise missing analysis

Importance of Understanding Missing Patterns

1. Helps Select the Right Imputation Technique

- Different missing patterns require different treatments. For example, MCAR can be handled with simple imputation, while MNAR requires advanced or domain-based approaches.

2. Prevents Bias in Analysis

- Ignoring missing patterns may introduce bias, leading to incorrect conclusions and unreliable models.

3. Improves Model Accuracy

- Proper handling of missing patterns ensures better data quality and improves the performance of statistical and machine learning models.

4. Aids in Data Quality Assessment

- Missing patterns often indicate issues in data collection processes, system failures, or survey design problems.

5. Supports Correct Statistical Inference

- Understanding whether data is MCAR, MAR, or MNAR is crucial for valid hypothesis testing and confidence estimation.

HANDLING NON NUMERICAL DATA IN MISSING PLACES

- Non-numerical data (also called categorical or qualitative data) includes values such as gender, department, status, city, or type. Missing values in such data require special handling because numerical imputation methods (mean/median) cannot be applied.

Causes of Missing Non-Numerical Data

- Non-response in surveys
- Data entry omissions
- Incomplete forms
- System or database errors

Techniques for Handling Missing Non-Numerical Data

1. Mode Imputation

- Missing values are replaced with the most frequently occurring category.
- Example:
Gender → Male, Female, Male, Missing → Male

Advantages:

- Simple and fast
- Maintains valid categories

Limitations:

- Can introduce bias if one category dominates

2. Constant or Placeholder Imputation

- Missing values are replaced with a special label such as “Unknown”, “Not Available”, or “Other”.
- Example:
Department → IT, CSE, Missing → Unknown

Advantages:

- Preserves information about missingness
- Useful when missing values are meaningful

Limitations:

- Adds an artificial category

3. Predictive (Model-Based) Imputation

- Machine learning models (e.g., decision trees, classification models) predict missing categories based on other variables.

Advantages:

- Uses relationships in data
- More accurate than simple methods

Limitations:

- Complex and computationally expensive

4. K-Nearest Neighbors (KNN) Imputation

- Missing categorical values are filled based on the most common category among similar records.

Advantages:

- Pattern-based
- Preserves data structure

Limitations:

- Sensitive to distance measure
- Slower for large datasets

5. Deletion of Records (Last Option)

- Records with missing non-numerical values are removed if they are very few and not important.

Limitation:

- Loss of data and information